

#Research Methodology

Unit - 1

Process:

- Observation
- Question
- Hypothesis
- Experiment
- Analysis
- Conclusion

Systematic, logical
& unbiased
process

Characteristics

- Systematic
- Critical
- Replicable
- Empirical
- Objective

Basic research = seeks knowledge for its own sake without immediate application (scientific principles)

Applied research = focuses on solving specific, practical problems (usable products) in industry

Qualitative research = focuses on understanding concepts through words & meanings

Quantitative research = focuses on measuring variables & testing hypothesis numerically

Exploratory research = investigates problems that are not clearly defined (used when problem scope is vague or new)

Problem formulation
 Literature review
 Research design
 Data collection
 Data analysis

Primary sources
 (surveys, direct observations)
 Secondary sources
 (analyzing existing datasets)

Tests — ANOVA
 — t-test
 Tools — SPSS
 — Python
 — R

Primary = high reliability, allows verification
 Secondary = bias (wider view but lack experimental details)

Research design = guides entire data collection process
 (framework)

Controls extraneous variables that could distort the relationship between variables

Literature reviews Expert interviews Pilot studies
 Types — Exploratory = problem that is not clearly defined yet
 Surveys — Descriptive = characteristics of phenomenon
 Case studies — Causal = cause & effect relationships between (experimental) specific variables

Exploratory = flexible & qualitative } large samples
 Descriptive = structured & quantitative }
 Experimental = highly controlled; small samples

Characteristics of a good research problem -

- Clear
- Specific
- Unambiguous

Sources of research problems:

- Literature reviews
- Theoretical inconsistencies in existing models
- Gov reports & policy changes
- Consultation with industry experts

Research gaps:

- ~~14~~ Population gaps = when a specific group has not been studied
- Knowledge gaps = ~~exist~~ when relationship between variables is not yet proven
- Practical application gaps = when theory exist but has not been implemented

Formulating research question:

PICO framework

- ~~Population~~ Population
- Intervention
- Comparison
- Outcome for clarity

Question should be open ended and narrow scope

Feasibility Criteria for problems -

- Time
- Financial
- Technical
- Data
- Ethical

Literature review = evaluates existing research to establish current state of knowledge
Avoid duplicating work & identifies gaps

Monographs = exhaustive scholarly book written by one author on a single specialized topic
Provide deep theoretical context
Often include extensive bibliographies

Patents = detailed technical solutions that may not be published in academic journals
Reveal practical limitations & engineering constraints of current industrial technologies
Patent classifications (IPC codes)
Organize technical knowledge

Synthesis = involves combining insights from academic reviews with practical data from patents

Types of Scholarly databases -

- IEEE Xplore = provides access to electrical engineering & C.S journals
- ScienceDirect = offers extensive coverage of physical sciences & engineering topics
- Scopus & web of science = multidisciplinary databases for citation analysis.
- Google scholar = acts as a broad search engine for academic literature
- Engineering Village = integrates multiple databases for comprehensive engineering searches

Advanced search operators -

- Quotation marks = for exact phrase matching
- AND operator
- OR operator
- NOT operator
- Parenthesis to group terms & control search logic order

Evaluating source credibility -

- Check if journal is peer reviewed before citing
- Verify the author's affiliation & expertise
- Examine publication date & look for high citation counts
- Avoid predatory journals that lack review

Search Strategy

- Query = 'solar energy' AND '~~the~~ lithium' NOT 'lead'
- Filter = Peer reviewed only, English language
- Gap identified
- Result
- Analysis

Types of Gaps in Literature -

- Empirical = when data is missing for a specific Population or Context
- Theoretical = when existing theories fail to explain new observed phenomena
- Methodological = when current research methods are insufficient
- Knowledge = where no research has been conducted at all
- Practical = highlight the disconnect between research findings

Strategies for identifying gaps -

- Read recent review articles
- Look for contradictory results
- Check limitations section of papers
- Search for studies that are outdated
- Analyze the scope of existing work

Questionnaires = structured sets of questions distributed to a large sample group
Ensure standardization but may suffer from low response rates

Interviews = direct verbal interaction to gather in depth qualitative insights
Allow for clarification but are time consuming & prone to interviewer bias

Sampling techniques for data gathering —
Ensures every member of the population has a known chance of selection

- Non probability sampling = relies on researcher judgement or convenience
- Simple random sampling = eliminates selection bias
- Stratified sampling = divides the population into subgroups
- Convenience sampling = quick but leads to significant selection bias errors

Designing data collection instruments

- Instruments must be clear, concise & free from ambiguous or leading questions
- Validity (evaluation)
- Reliability (consistent results)
- Pilot testing = small trial run to identify flaws before full deployment
- Well designed instrument minimizes measurement error & ~~then~~ maximizes data quality

$$\text{Sample size } (n) = \frac{(Z^2 \times p \times (1-p))}{e^2}$$

for infinite populations

Z = Z score for corresponding to desired confidence level

p = estimated proportion of attribute present in the population

e = margin of error acceptable in the study results

Organization

Synthesis of Qualitative findings —

- Synthesis involves integrating findings from multiple sources to form unified wh
- Researchers identify recurring themes & patterns across different qualitative datasets
- Thematic analysis weaves findings into a coherent story

Synthesis of Quantitative findings —

- Aggregates numerical data to identify statistical trends
- Meta analysis statistically reveal the strength & direction of relationships
- Trend analysis track changes in data over time to predict future outcomes

Techniques for pattern identification —

- Coding assigns labels to data segments to categorize & organize information
- Matrix analysis compares data points across different variables simultaneously
- Constant comparison method requires categorizing by comparing new data
- Triangulation validates findings by cross checking data from multiple source

Purpose of Analysis -

- Data analysis = process of inspecting, cleaning & modeling data
- Transform raw observations into structured info
- Bridge between data collection & final research conclusions

Descriptive vs Inferential analysis -

- Descriptive = summarizes main features of a dataset
(mean, median, mode, SD)
- Inferential = uses sample data to make predictions
(hypothesis testing & confidence interval estimation)

Data coding & tabulation -

- Assigns numerical or symbolic labels to qualitative data categories
- Process converts unstructured responses into a format suitable
- Ensures consistency & reduces error

Data cleaning -

- Missing values
- Outliers
- Standardization

Interpretation -
Meaning of analyzed data in context

Logical steps

- Review
- Compare findings
- Identify patterns
- Deduce generalizations

Common interpretation errors -

- Avoids overgeneralizing results
- Do not confuse correlation with causation
- Prevent personal bias

Text

- Use clear, concise language to describe key findings
- Summarize statistical results in narrative form
- Avoid jargon

Visuals

- Use tables for precise numerical data & comparisons
- Employ graphs to show trends
- Ensure all visual elements have clear labels & legends